

## Theoretical Exam– English Language

12 January 2024, 14:30h

duration: 120 minutes

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- **Justify** all your answers in a simple, clear, succinct, and objective way.
  - Beside your name and student number, please also write your **Team number**.
  - Write the answers to questions **1.** and **2.** on the **second** sheet of paper, and the **remaining** ones on the **first** sheet.
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1. **[15%]** Regarding Git:

- a) Describe how your team managed code and documents using Git, namely explaining the workflow used, how files were created and distributed, how collaborative development functioned, and how you ensured the quality of software and text documents, etc.
- b) In what ways did Git have a positive impact on your team's work?
- c) Identify 2 specific challenges that you encountered and overcame.

2. **[10%]** How have you personally incorporated and applied the fundamental values of Scrum (courage, focus, commitment, respect, and openness) as a member of your team? Respond based on at least two specific situations where adherence to one or more of these values positively influenced the project outcome or team dynamics.

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3. **[10%]** About Traditional and Agile Development:

- a) Considering other projects that you have undertaken in other programming-related courses, namely:
  - Advanced Programming, or Web Developmentwhat would be the main differences in applying a traditional methodology versus an agile methodology in such projects?
- b) Regarding these projects, how would it be beneficial for the students to follow an agile methodology?

4. **[15%]** Relate User Stories, Tasks, Acceptance Criteria, Unit Tests, and Acceptance Tests using examples for their contents, based on a login functionality.

5. **[10%]** Provide an estimate for the date of delivery of the grades for this exam.

- a) You should explain how you arrived at this estimate.
- b) Present the assumptions used to obtain this value.
- c) If you average the values obtained by each member of your group, would the result be more or less accurate than estimating all together?

Note: The estimated value by itself will not contribute to the assessment of this question.

6. [15%] The threshold of success of a web application is as follows:

- **Threshold of success:** “At the end of 1 year of usage, sensitive user data should not have been accessed by any hacker.”

Present the Risk, in the format of a **Risk Statement**, as presented in the classes, and provide a risk mitigation plan consisting of:

- An avoidance strategy;
- A minimization plan;
- A contingency plan.

7. [10%] Imagine that during the development of your project, a problem occurred in a feature, near the end of the sprint, preventing the team from presenting the demo as agreed with the client. Present two possible approaches to tackle this problem, emphasizing the advantages of each one relative to the other.

8. [15%] Observe the process in the following figure and explain:

- What is the objective
- What are the inputs
- What are the outputs
- What are the steps

Be clear in explaining the steps.

